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RESEARCH ARTICLE

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Towards a cutting-edge metallization process for silicon heterojunction solar cells with very low silver laydown

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Abstract

Within this work, we investigate the potential to optimize the screen-printed front side metallization of silicon heterojunction (SHJ) solar cells. Three iterative experiments are conducted to evaluate the impact of the utilized fine mesh screen configurations and grid layout adaption (finger pitch) for the front side metallization on silver laydown and electrical performance of the solar cells. With respect to the screen configuration, we compare the performance of a fine-mesh knotless screen to a conventionally angled screen demonstrating an additional gain of $\Delta\eta = +0.1\%_{abs}$ due to reduced shading losses. Additionally, a grid layout is improved by increasing the number of contact fingers from 120 to 156. Furthermore, the current possibility to push the fine-line printing process for low-temperature pastes to the limit is investigated by reducing the nominal finger width w_n to 20, 18, and 15 µm. It is shown that even the smallest nominal width of $w_n = 15$ µm can be printed with high quality, leading to an additional efficiency gain of $\Delta\eta = +0.15\%_{abs}$ as well as a reduction of silver paste laydown by -5 mg. Finally, a batch of champion cells is fabricated by applying the findings of the previous experiments, which results in a maximum efficiency of $\eta_{max} = 23.2\%$. Compared to the reference group without optimization, this corresponds to a gain of $\Delta\eta = +0.17\%_{abs}$, which comes along with an additional decrease of the silver paste laydown by approximately -2 mg. This emphasizes the significance of consistent optimization of the screen-printing process in terms of cell performance and resource utilization for SHJ solar cells.

KEYWORDS
fine-line metallization, screen printing, silicon heterojunction solar cells, silver laydown

1 | INTRODUCTION

High-efficiency solar cell concepts with passivating contacts[1] have gained a considerable share in the global industrial PV production and will increasingly displace the currently dominating PERC (passivating emitter and rear contact) cell concept.[2] Among various industrially fabricated high-efficiency cell concepts, silicon heterojunction (SHJ) solar cells[1,3] represent a particularly promising approach with recently demonstrated record conversion efficiencies up to $\eta = 26.81\%$.[4] However, several challenges must be addressed to ensure the long-term cost competitiveness of industrially fabricated SHJ solar cells and modules. One of the primary challenges for the coming years is

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the increasing shortage of critical resources, namely, silver, indium, and bismuth.[5] Based on the projected growth of installed PV capacity in the coming years, the silver consumption of the PV industry could increase to up to 90% of the global annual production volume until 2030.[5] The need to reduce silver laydown is particularly urgent for SHJ solar cells as silver paste is usually applied on the front and rear sides of typical cell architectures. Various promising approaches for a substantial reduction of the silver consumption per W_p are currently investigated, namely, fine line screen printing with very low silver laydown,[6,7] replacement of conventional silver paste by silver-coated copper paste[8] or reactive silver ink[9] or new metallization approaches with very low silver laydown like FlexTrail,[10] rotary screen printing,[11] and parallel dispensing.[12] Promising results have also been demonstrated using electroplating of copper (Cu) electrodes on SHJ solar cells.[13]

Focusing on the reduction of silver laydown using fine line screen printing, it is crucial to minimize the finger width of the printed contact grid while maintaining a sufficiently low grid resistance R_{Grid} for the intended interconnection concept.[14] The maximum tolerable mean contact finger resistance R_l for a given series resistance contribution Δr_s of the front and rear side grid strongly depends on the selected interconnection approach and the resulting optimal grid layout. Using a cell interconnection concept with multiple wires (multi-wire interconnection[15–17]) as shown in Figure 1 instead of the traditional interconnection via soldered ribbons on printed busbars substantially reduces the finger segment length in between the wires (thus substantially increasing the tolerance regarding r_s-losses due to the lateral finger resistance R_L[15,18]) and in parallel increases the j_{sc} by reduced shading losses on module level.[17,19,20]

Thus, multi-wire concepts like SmartWire Interconnection Technology (SWCT)[16,21] can increase the efficiency on module level on the one hand and in parallel substantially decrease the silver usage on cell level on the other hand.[17] Using fine-mesh screens to print very fine contacts,[6,7] even the predicted goal to reduce the silver consumption below 5 mg/W_p[5] seems to be within reach.[7] Yet, while such interconnection concepts can help to drastically reduce the silver usage on the one hand, the necessity to use low temperature alloys that usually include rare materials like indium or bismuth does pose other challenges with respect to sustainability.

However, achieving this goal requires a reliable and industrially applicable fine-line printing process on the front and rear side of SHJ solar cells which is even more challenging with low-temperature (LT) pastes. Reliably printing fine-line contacts below 20 µm or less requires high-performance materials (screen and paste). Furthermore, the printing conditions (printing speed and pressure) can have a significant impact on the metallization result.[22,23] Furthermore, the grid layout must be optimally designed to obtain an optimal balance regarding shading losses (finger width and number of contacts) and series resistance losses (grid resistance and number of contacts). Within this work, we evaluate several important parameters for a cutting-edge fine-line printing process including screen configuration (standard angled screen vs. knotless screen[6,24]), grid layout design (number of contact fingers), and nominal finger width w_n (target finger width in the grid layout).

2 | RESULTS AND DISCUSSION

2.1 | Comparison of knotless versus standard screens

The usage of so-called "knotless" screens with a mesh angle of 0° has proven to be beneficial for the reliable printing of narrow and uniform contact fingers in various recent studies.[6,24,25] However, it is important to note that these studies conducted the experiments using high-temperature sintering silver paste on PERC solar cells. To assess the impact of the screen mesh angle on the front side metallization of SHJ solar cells, the first experiment focuses on the comparison of an advanced knotless fine-mesh screen (520 × 11 × 0°) to a conventional angled fine-mesh screen (520 × 11 × 22.5°). The front side metallization is screen printed using both screen types with the same printing conditions as described in Section 4 and a printing/flooding speed of $v_{print}/v_{flood} = 300$ mm/s. Both types of screens obtained excellent printing results with respect to the front side metallization of the cells. Table 1 shows the results of the finger geometry analysis based on confocal laser scanning microscopy (CLSM) measurement and image analysis of selected contact fingers.

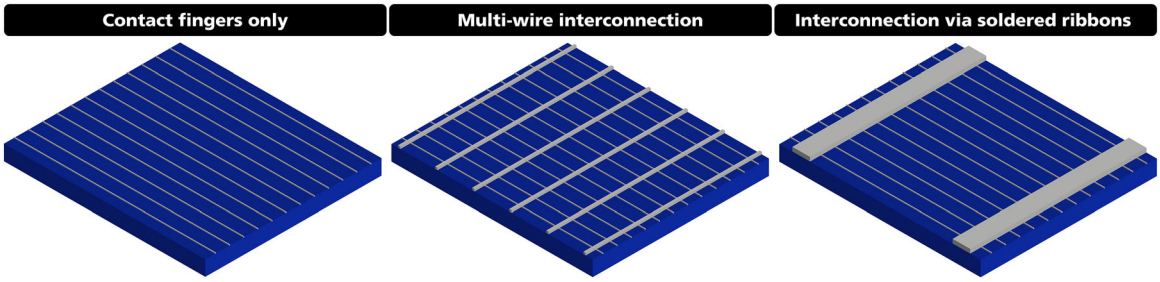

FIGURE 1 Schematic illustration of different interconnection technologies. Left side: metallized busbarless solar cell with screen printed front side contacts. Middle: busbarless solar cell interconnected via multiple wires. Right: interconnection via soldered ribbons on screen printed busbars. The considerably smaller finger segment length between the wires for a multi-wire interconnection is clearly visible.

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TABLE 1 Mean values and 95% confidence interval of finger geometry parameters of all groups based on CLSM measurement and image analysis

Group	Screen type	Nom. finger width (front side) w_n (μm)	Avg. shading finger width w_f (μm)	Avg. finger height h_f (μm)	Avg. cross-section area A_f (μm)
1	520 × 11 × 0°	20	27 ± 2.2	7 ± 0.6	94 ± 9.0
2	520 × 11 × 22.5°	20	28 ± 2.5	7 ± 0.8	100 ± 14.3

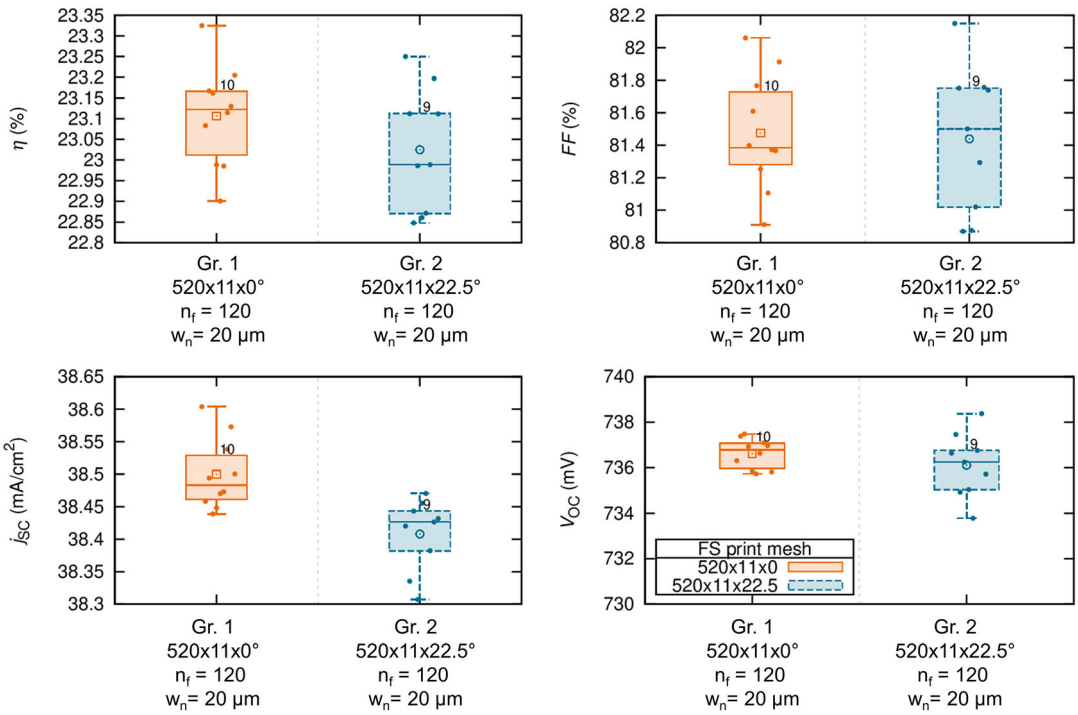

FIGURE 2 I-V results of Experiment 1 using a knotless versus conventionally angled screen for the front side metallization. The knotless screen proved to be particularly beneficial to increase the j_{sc} by approximately +0.5 mA/cm^2 (reduced finger shading width w_f), while the average fill factor FF is on the same level (with a greater dispersion of the measurement values for Group 2). These effects result in a total efficiency increase of $\Delta\eta = 0.1\%_{abs}$.

The results show that a narrower mean finger width ($\Delta w_f \approx 1.3$ μm) can be obtained using the knotless instead of a standard-angled screen. The average finger height h_f of both groups is comparable resulting in a slightly lower average finger cross-section area A_f for Group 1. The front side grids printed with the knotless screen also show a better uniformity, indicated by the lower variation of w_f, h_f, and A_f. This is particularly important when printing very narrow fingers as pronounced variations of the finger uniformity (i.e., constrictions and interruptions) often negatively affect the lateral resistance and thus the FF of the solar cells. Knotless screens are thus particularly beneficial when printing grid layouts with a very narrow nominal finger width w_n. Figure 2 illustrates the results of the I-V measurement of the SHJ solar cells printed with both types of screens.

The I-V results indicate that the knotless screen proved to be beneficial with respect to an increased short-circuit current density j_{sc} by +0.056 mA/cm^2, which is attributed to the smaller finger shading width w_f. Both groups achieve an average FF at the same level, with Group 1 showing a smaller dispersion of the measurement values due to the better uniformity of the printed contact fingers.

In summary, these effects result in a total efficiency increase of $\Delta\eta = 0.1\%_{abs}$, which proves the advantage of knotless fine-mesh screens with respect to an increased performance of the front side metallization and thus an overall efficiency increase.

2.2 | Variation of finger pitch

Printing narrower contact fingers requires a parallel optimization of the metallization grid layout to find an optimal trade-off between series resistance losses (due to a typically increased average finger contact resistance of shallower/smaller contacts) and j_{sc} gains (less shading due to a narrower average shading finger width w_f). Thus, a variation of the finger pitch of the front side grid layout is conducted in the second experiment. Using the beneficial knotless screen configuration (520 × 11 × 0°) from the first experiment, two busbarless grid

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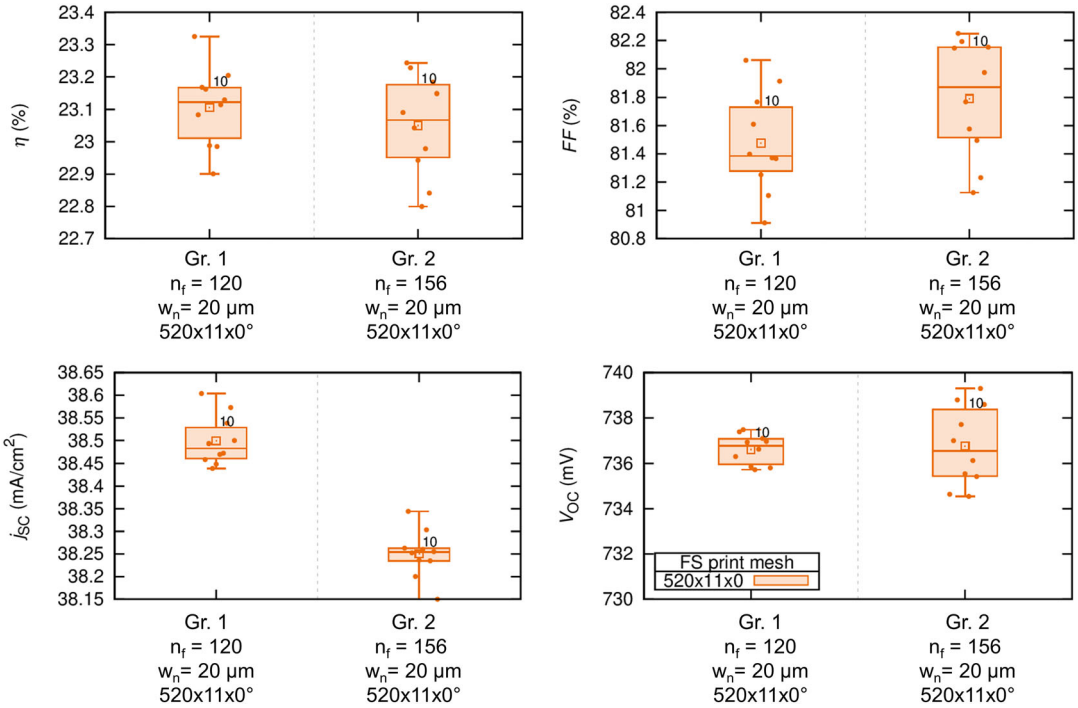

FIGURE 3 I-V results of Experiment 2 with varying finger pitch of the front side grid layout. Decreasing the finger pitch from 1.3 mm (120 fingers) to 1.0 mm (156 fingers) results in an increasing *FF* ($\Delta FF = +0.4\%_{abs}$) on the one hand and a decreasing j_{sc} ($\Delta j_{sc} \approx -0.25$ mA/cm^2) on the other hand. In total, a closely similar conversion efficiency for both groups is visible.

layout variants with a finger pitch of 1.3 mm (120 fingers) and 1.0 mm (156 fingers) are used for the front side metallization. By increasing the amount of contact fingers in the front side grid layout, a decreasing j_{sc} due to higher shading losses (increased total area of metallization) can be expected. On the other side, the *FF* should increase due to a reduced grid resistance in between two wires used for the interconnection of the cells (more fingers for current transport available) and the smaller finger pitch that results in reduced losses in lateral transport (base, TCO) to the grid fingers. Figure 3 shows the I-V results of both groups. Consistent with expectations, reducing the finger pitch from 1.3 mm (120 fingers) to 1.0 mm (156 fingers) results in an increased *FF* ($\Delta FF = +0.4\%_{abs}$), while the short-circuit current density j_{sc} ($\Delta j_{sc} \approx -0.25$ mA/cm^2) decreases due to increased shading. Both effects offset each other to a large extent in this specific case, resulting in a comparable conversion efficiency for both groups. With respect to silver consumption, reducing the finger pitch increases the Ag paste laydown from 19 mg (1.3 mm pitch, 120 fingers) to 21 mg (1.0 mm pitch, 156 fingers).

2.3 | Variation of nominal finger width

Reducing the printed finger width of the front (and rear) side metallization is beneficial with respect to an increased efficiency (reduced shading losses) and a substantial silver reduction. By adequately reducing the finger pitch (increasing the number of contact fingers in the grid layout), grid-related series resistance losses can be avoided, yet at the expense of an increased Ag laydown. While very narrow contact fingers down to a width of ~20 µm and less[6,25] have been successfully demonstrated, it is far more challenging to reliably print such narrow finger widths using low-temperature sintering silver paste. To date, typical finger widths using LT silver paste on industrially fabricated SHJ solar cells are in the range of 30–40 µm. A systematic evaluation of the current limit with respect to a decreasing finger width using LT silver paste is thus highly interesting and relevant for a further improvement of SHJ solar cells and a substantial reduction of silver usage. To identify the current limit, three advanced fine-mesh screens (520 × 11 × 0°) with varying nominal finger width w_n (20, 18, and 15 µm) are fabricated and used for printing the front side metallization. Table 2 shows the average printed finger geometry as well as the silver paste laydown of the three screen variations after curing. Reducing the nominal finger width w_n obviously affects the finger geometry as well as the silver paste laydown on the front side, which is clearly visible when comparing the results shown in Table 2 and Figure 4.

Surprisingly, it is possible to print even the grid layout with a very small nominal finger width of $w_n = 15$ µm reliably and with good visual quality using the given high-performance screen specification and LT silver paste. This enables a reduction of the shading finger width from $w_{f1} = 26$ µm (nom. finger width $w_n = 20$ µm) to $w_{f3} = 23$ µm (nom. finger width $w_n = 15$ µm). The silver laydown on the front side can be consequently reduced from $m_{Ag1} = 21$ mg to $m_{Ag3} = 16$ mg.

Considering the I-V results shown in Figure 5, an increase of the short-circuit current by approximately $\Delta j_{sc} \approx +0.1$ mA/cm^2 due to the

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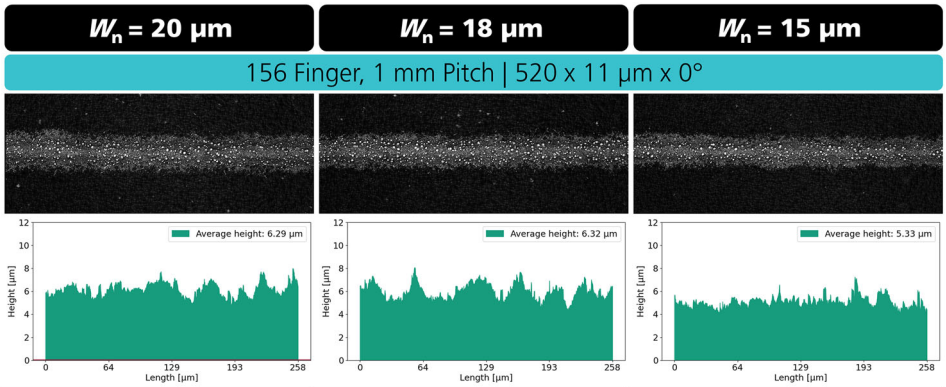
TABLE 2 Mean values and 95% confidence interval of finger geometry parameters of all groups based on CLSM measurement and image analysis and average silver paste laydown

Group	Screen type	w_n (µm)	Avg. finger width w_f (µm)	Avg. finger height h_f (µm)	Avg. cross-section area A_f (µm)	Avg. silver paste laydown (mg)
1	520 × 11 × 0°	20	27 ± 2.2	7 ± 0.6	94 ± 9.0	21
2	520 × 11 × 0°	18	26 ± 2.2	6 ± 0.6	86 ± 8.1	20
3	520 × 11 × 0°	15	23 ± 2.2	6 ± 0.5	71 ± 6.8	16



	W_n = 20 µm	W_n = 18 µm	W_n = 15 µm
$W_{f,core}$	20 µm	19 µm	17 µm
$W_{f,shading}$	26 µm	25 µm	23 µm
H_f	7 µm	6 µm	5 µm

FIGURE 4 CLSM images and image analysis results of the finger geometry using screens with a nominal finger width of $w_{n1} = 20$, $w_{n2} = 18$, and $w_{n3} = 15$ µm. All three screens could be printed with excellent and reproducible quality leading to a minimum average core finger width of $w_c = (17 \pm 1)$ µm and a shading finger width of $w_f = (23 \pm 1)$ µm using the screen width a nominal finger width of $w_n = 15$ µm.

FIGURE 5 I-V results of Experiment 3 using three screen variations with varying nominal finger width w_n (20, 18, and 15 µm). It is clearly visible that the reduction of the w_n from 20 to 15 µm particularly leads to a j_{sc} gain (due to the smaller average finger shading width and thus reduced shading losses), resulting in an average efficiency gain of $\Delta\eta = +0.14\%_{abs}$.

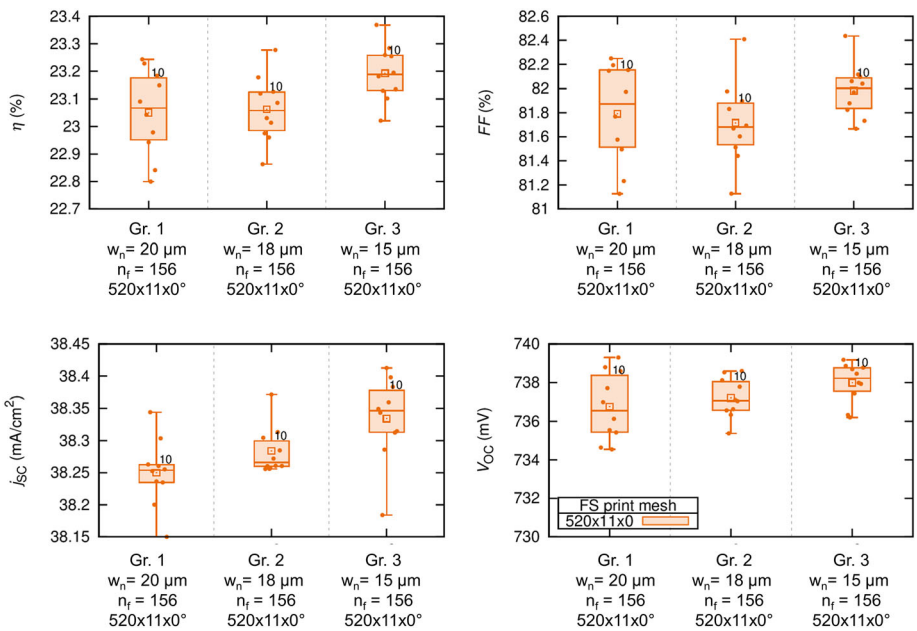


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TABLE 3 I-V-results of the bifacial heterojunction solar cells measured with a GridTouch® unit in an industrial cell tester as well as silver laydown values on the front side

Group	Screen type	w_n (µm)	p_f (mm)	Avg/best	J_{sc} (mA/cm^2)	V_{oc} (mV)	*FF* (%)	η (%)	Avg. silver laydown (mg)
1	520 × 11 × 22.5°	20	1.3	Avg	38.41	736.1	81.44	23.02	18.7
			1.3	Best cell	38.42	736.6	82.15	23.25	-
2	520 × 11 × 0°	15	1.0	Avg.	38.33	738.0	81.98	23.19	16.3
			1.0	Best cell	38.35	739.2	82.44	23.37	-

smaller contact fingers and thus less shading losses is clearly visible. The average FF also shows minor variations when comparing all three groups. It is likely that these minor variations do not result from the metallization but from variations within the used precursor material (as indicated by pFF variations). Group 3 achieves an average efficiency gain of approximately $\Delta\eta = +\ 0.14\%_{abs}$ compared to Group 1. Considering the additionally reduced silver paste laydown, it can be clearly concluded that reducing the nominal finger width can have a beneficial effect on both conversion efficiency and silver reduction provided that an optimal screen configuration is used.

2.4 | Fabrication of SHJ solar cells with optimized parameters

Based on the findings of the previous experiments, high-performance SHJ solar cells are manufactured using the knotless screen (520 × 11 × 0°) with a finger pitch of $p_f = 1.0$ mm and a nominal finger width $w_n = 15$ µm. The front side metallization is carried out at a printing speed of $v_{print}/v_{flood} = 300$ mm/s. The I-V results (measured with GridTouch®) of both groups are shown in Table 3. Optimized Group 2 achieved an average efficiency gain of $\Delta\eta = +0.17\%_{abs}$ compared to non-optimized Group 1, representing a substantial increase. Furthermore, the average silver laydown of Group 2 could be decreased by ~2 mg compared to Group 1. This indicates a clear advantage of the utilized knotless fine-mesh screen in combination with an adapted grid layout.

3 | SUMMARY AND CONCLUSION

Within this work, three iterative experiments are conducted to assess and optimize the fine-line screen-printing process of SHJ solar cells. Within the first experiment, a high-performance knotless screen is compared to a conventionally angled fine-mesh screen regarding the metallization result. The knotless screen proved to be beneficial with respect to an increased short-circuit current density j_{sc} by approximately +0.5 mA/cm^2 due to a reduced finger shading width w_f. However, the fill factor FF is also slightly lower, leading to an overall efficiency gain of $\Delta\eta = +0.14\%_{abs}$ for the knotless screen.

Within the second experiment, the finger pitch p_f is decreased by using 156 instead of 120 finger in the front side grid layout. As expected, the reduced finger pitch leads to an increasing *FF* ($\Delta FF = +0.4\%_{abs}$) due to additional current paths in between the wires (and shorter distance in the TCO) and a decreasing j_{sc} ($\Delta j_{sc} \approx -0.25$ mA/cm^2) due to additional shading. In this particular case, both effects more or less balance each other out, resulting in a closely similar conversion efficiency for both groups yet at the expense of an increased silver paste laydown by ~2 mg. However, it is very likely that a further decrease of the printed finger width w_f combined with a reduced finger pitch (higher amount of fingers) will have a positive effect on the overall conversion efficiency.

The third experiment evaluates the limit of fine-line printing under the given conditions by varying the nominal finger width between 20, 18, and 15 µm. The results show that even the smallest nominal finger, width $w_n = 15$ µm can be printed reliably and with good quality leading to an increase of the short-circuit current density by approximately $\Delta j_{sc} \approx +0.1$ mA/cm^2 compared to a broader nominal finger width of $w_n = 20$ µm. This leads to a substantial efficiency gain of approximately $\Delta\eta = 0.14\%_{abs}$. At the same time, the silver laydown can be reduced by −5 mg on the front side, which is a highly beneficial effect for industrial production.

Finally, SHJ solar cells are fabricated with optimized conditions based on the previous findings. The best group obtained an average conversion efficiency of $\eta_{max} = 23.2\%$, which corresponds to a gain of $\Delta\eta = +0.17\%_{abs}$ compared to the reference cells without the described optimization. Furthermore, the silver paste laydown of this group could be reduced by ~2 mg.

In summary, we demonstrate several approaches to optimize the front side metallization and in parallel substantially reduce the silver laydown of SHJ solar cells, leading to a reduced silver laydown and an overall efficiency increase of $0.17\%_{abs}$.

4 | EXPERIMENTAL METHODS

4.1 | Experimental setup

Within this work, three experiments have been conducted to evaluate and optimize the fine-line screen-printing process for SHJ solar cells. An actual low-temperature silver paste from *Namics Corporation* and a fine-mesh knotless screen (520 × 11 × 0°) with a nominal finger width of $w_n = 20$ µm on the front side and $w_n = 40$ µm on the rear side is used for the metallization process. All cells are metallized on an ASYS screen-printing line at Fraunhofer ISE PVTEC. Subsequently, the cells are dried and cured at a curing time of $t_{cure} = 1$ min and a curing temperature of $T_{cure} = 220$°C.

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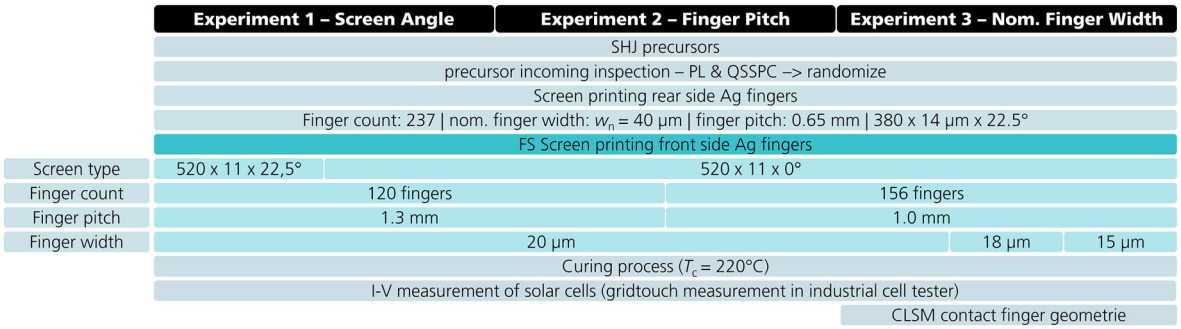

FIGURE 6 Experimental setup and process steps for Experiments 1–3

TABLE 4 Overview of the various experiments conducted within this work

Exp. No.	Variation	Value	Cells per group
1	Screen: mesh angle	22.5° and 0°	10
2	Grid layout: No. of contact fingers n_f	120 to 156 fingers	10
3	Grid layout: Nom. finger width w_f	20-18-15 µm	10

Figure 6 illustrates the experimental plan for Experiments 1–3. Within Experiment 1, the screen specification (mesh angle) to apply the fine-line front side metallization is varied by comparing the performance of a fine-mesh screen with a mesh count of 520 wires/in., a wire thickness of $d_{wire} = 11$ µm and a mesh angle of $\varphi = 22.5°$ (520 × 11 × 22.5°) to a fine-mesh knotless screen (520 × 11 × 0°). The same type of paste and comparable printing parameters are used for the print run.

Experiment 2 focuses on the impact of the busbarless grid layout by increasing the number of contact fingers from 120 to 156 fingers or, respectively, decreasing the finger pitch from $p_{f1} = 1.3$ mm to $p_{f2} = 1.0$ mm in order to optimize the amount of current paths in between to wires/busbars. Within Experiment 3, the nominal finger width w_n is varied in three stages (20, 18, and 15 µm) to identify the minimal printable finger width w_f with the applied paste, screen, and printing parameters. An overview of the experimental groups is presented in Table 4.

4.2 | Materials

Industrially prefabricated SHJ cells (157.35 mm × 157.35 mm) with transparent conductive oxide (TCO) on both sides but without metallization have been used for the experiment. The used material can obtain a maximum efficiency potential in the range of 23–24%.

4.3 | Variation of fine-mesh screens

The front side metallization of the SHJ solar cells is realized using fine mesh screens with a mesh count of $m_c = 520$ wires/in. and a nominal wire thickness of $d_{wire} = 11$ µm. All screens are equipped with a high-performance emulsion with a nominal emulsion over mesh (EOM) thickness of $t_{EOM} = 13$ µm. To compare the impact of the mesh angle on the front side metallization and the cell performance, two screen variants have been compared: on the one hand, a fine-mesh screen with a standard mesh angle of $\varphi_1 = 22.5°$ and, on the other hand, a so-called "knotless screen" with a mesh angle of $\varphi_2 = 0°$. Due to the prevention of wire intersections (knots) in the finger channels, this type of screen typically enables a substantially better uniformity of the contact fingers (considerably less "mesh marks"[24]) and thus a lower mean lateral finger resistance R_{LINE} compared to standard screens with a mesh angle of $\varphi_1 = 22.5°$.[6,24,26] All screens are printed with the same printing conditions. An overview of the utilized screens is shown in Table 5.

4.4 | Grid layout

Various grid layout designs are used for Experiments 1 to 4. The individual grid layouts and screen specifications for the different test runs are shown in Table 5.

4.5 | Screen-printing process

The screen-printing process is carried out on a ASYS screen-printing line at Fraunhofer ISE PVTEC. To compare the performance of the different screen variants, the printing process for Experiments 2 and 3 is performed with identical printing parameters, namely, a printing speed of $v_{print} = 300$ mm/s and a squeegee pressure of $p_{print} = 75$ N.

4.6 | Drying and curing parameters

All cells are dried and cured in an inline belt drying furnace at a set temperature of $T_c = 220°$ for $t_c = 1$ min.

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TABLE 5 Overview of the screen and grid layout parameters used for Experiments 1–3

Exp. No.	Front/ rear side	Mesh count (1/in.)	Wire diameter (µm)	Mesh angle φ (°)	Nom. Finger width w_n (µm)	Number of fingers n_f (–)	Finger pitch (mm)	Number of busbars (–)
1	Front	520	11	22.5	20	120	1.3	-
	Front	520	11	0	20	120	1.3	-
	Rear	520	11	0	24	196	0.8	-
2	Front	520	11	0	20	120	1.3	-
	Front	520	11	0	20	156	1.0	
	Rear	520	11	0	24	196	0.8	-
3	Front	520	11	0	20	156	1.0	-
	Front	520	11	0	18	156	1.0	-
	Front	520	11	0	15	156	1.0	-
	Rear	520	11	0	24	196	0.8	-

4.7 | I-V-parameters and grid resistance measurement

The metallized and cured cells are measured using an industrial cell tester with GridTouch®-unit to evaluate the I-V-results. The GridTouch®-unit is equipped with 30 wires for current and 5 wires for voltage measurement. The lateral resistance R_{Grid} and, respectively, the average lateral finger resistance R_{LINE} of the grid cannot be measured accurately with GridTouch®. Thus, a separate dual printing step to apply five busbars ($w_b = 800$ µm) on the front and rear side using low temperature paste is applied. Subsequently, the SHJ cells with imprinted busbars are measured again on the cell tester using standard measuring bars for H-pattern grids to precisely determine the lateral grid resistance R_{Grid} on the front and rear side. The mean lateral finger resistance R_{LINE} per contact finger can be calculated by Equation (1) using measured busbar to busbar resistance $R_{BB\text{-}BB}$, the number of contact fingers n_f, and the finger segment length l_f between two busbars[24]:

$$R_{LINE\emptyset} = \frac{n_f \cdot R_{BB-BB}}{l_f} \text{for rBB} < < \text{rf}. \quad (1)$$

4.8 | Finger geometry measurement

An in-depth analysis of the finger geometry after printing and curing is conducted. Twnety-five measurements per group are carried out on the front side grid using an Olympus LEXT confocal laser-scanning microscope (CLSM) with 500X magnification. All images are statistically analyzed using the Fraunhofer ISE image analysis software *Dash*.[6] This software allows a statistical determination of all relevant geometry parameters including average values for the core finger width w_c, the shading-relevant finger width w_f, finger height h_f, and finger cross-section area A_f.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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REFERENCES

1. Hermle M, Feldmann F, Bivour M, Goldschmidt JC, Glunz SW. Passivating contacts and tandem concepts: approaches for the highest silicon-based solar cell efficiencies. *Appl. Phys. Rev.* 2020;7(2):21305. doi:10.1063/1.5139202
2. ITRPV, International Technology Roadmap for Photovoltaics (ITRPV): Results 2021, thirteenth edition, 2022.
3. de Wolf S, Descoeudres A, Holman ZC, Ballif C. High-efficiency silicon heterojunction solar cells. A review. *Green*. 2012;2(1):7-24. doi:10.1515/green-2011-0018
4. Lin H, Yang M, Ru X, et al. Silicon heterojunction solar cells with up to 26.81% efficiency achieved by electrically optimized nanocrystalline-silicon hole contact layers. *Nat. Energy*. 2023;8(8):789-799. doi:10.1038/s41560-023-01255-2
5. Zhang Y, Kim M, Wang L, Verlinden P, Hallam B. Design considerations for multi-terawatt scale manufacturing of existing and future photovoltaic technologies: challenges and opportunities related to silver, indium and bismuth consumption. *Energ. Environ. Sci.* 2021;14(11):5587-5610. doi:10.1039/D1EE01814K
6. Wenzel T, Lorenz A, Lohmüller E, et al. Progress with screen printed metallization of silicon solar cells - towards 20 µm line width and 20 mg silver laydown for PERC front side contacts. *Sol. Energy Mater.* 2022;244:111804. doi:10.1016/j.solmat.2022.111804

1099159x, 2024, 10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/pip.3808 by Fraunhofer IRB, Wiley Online Library on [09/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

7. Pingel S, Wenzel T, Göttlicher N, et al. Progress on the reduction of silver consumption in metallization of SHJ solar cells. *Sol. Energy Mater*. 2024;265:112620. doi:10.1016/j.solmat.2023.112620
8. Nam J-B, Jang YR, Hwang YT, Kim HH, Jung IH, Kim HS. Intense pulsed light sintering of screen-printed paste electrode on silicon solar cell for high throughput and cost-effective low temperature metallization. *Int. J. Precis. Eng. Manuf. Green Technol.* 2021;9(2):523-535. doi:10.1007/s40684-021-00346-3
9. DiGregorio SJ, Martinez-Szewczyk M, Raikar S, Bertoni MI, Hildreth OJ. Investigation of reactive silver ink formula for reduced silver consumption in silicon heterojunction metallization. *ACS Appl. Energy Mater*. 2023;6(5):2747-2757. doi:10.1021/acsaem.2c03503
10. Schube J, Jahn M, Pingel S, et al. FlexTrail printing as direct metallization with low silver consumption for silicon heterojunction solar cells: evaluation of solar cell and module performance. *Energ. Technol.* 2022;10(12):ente.202200702. doi:10.1002/ente.202270124
11. Lorenz A, Klawitter M, Linse M, et al. Rotary screen printed metallization of heterojunction solar cells: toward high-throughput production with very low silver laydown. *Energ. Technol.* 2022;10(8):2200377. doi:10.1002/ente.202200377
12. Gensowski K, Jimenez A, Tepner S, et al. Dispensing of low-temperature silver pastes. In: *AIP Conference Proceedings*. Vol.2367, No. 1. AIP Publishing LLC; 2021:20007.
13. Yu J, Bai Y, Li J, et al. Process challenges of high-performance silicon heterojunction solar cells with copper electrodes. *Sol. Energy Mater*. 2023;250:112057. doi:10.1016/j.solmat.2022.112057
14. Tepner S, Lorenz A. Printing technologies for silicon solar cell metallization: a comprehensive review. *Progr. Photovolt. Res. Appl.* 2023;31(6):557-590. doi:10.1002/pip.3674
15. Schneider A, Rubin L, Rubin G. "Solar cell efficiency improvement by new metallization techniques—the Day4 electrode concept," In: 2006 IEEE 4th WCPEC, Hawaii, pp. 1095–1098, 2006.
16. Söderström T, Papet P, Ufheil J. "Smart wire connection technology," In: Proc. of the 28th Photovoltaic Solar Energy Conference (EUPVSEC) Paris, France, pp. 495–499, 2013.
17. Braun S, Nissler R, Ebert C, Habermann D, Hahn G. High efficiency multi-busbar solar cells and modules. *IEEE J. Photovolt*. 2014;4(1):148-153. doi:10.1109/JPHOTOV.2013.2286525
18. Walter J, Tranitz M, Volk M, Ebert C, Eitner U. Multi-wire interconnection of busbar-free solar cells. *Energy Procedia*. 2014;55:380-388. doi:10.1016/j.egypro.2014.08.109
19. Braun S, Micard G, Hahn G. Solar cell improvement by using a multi busbar design as front electrode. *Energy Procedia*. 2012;27:227-233. doi:10.1016/j.egypro.2012.07.056
20. Braun S, Hahn G, Nissler R, Pönisch C, Habermann D. Multi-busbar solar cells and modules: high efficiencies and low silver consumption. *Energy Procedia*. 2013;38:334-339. doi:10.1016/j.egypro.2013.07.286
21. Hermans JP, Papet P, Pacheco K, et al. "Advanced metallization concepts by inkjet printing," In: Proc. of the 29th Photovoltaic Solar Energy Conference (EUPVSEC) Amsterdam, Netherlands, pp. 518–522, 2014.
22. Tepner S, Wengenmeyr N, Linse M, Lorenz A, Pospischil M, Clement F. The link between ag-paste rheology and screen-printed solar cell metallization. *Adv. Mater. Technol.* 2020;5(10):2000654. doi:10.1002/admt.202000654
23. Erath D, Pingel S, Khotimah R, et al. Fast screen printing and curing process for silicon heterojunction solar cells. In: *AIP Conference Proceedings*. Vol.2367, No. 1. AIP Publishing LLC; 2021:20006.
24. Lorenz A, Linse M, Frintrup H, et al. "Screen printed thick film metallization of silicon solar cells—recent developments and future perspectives," In: Proc. of the 35th Photovoltaic Solar Energy Conference (EUPVSEC) Brussels, Belgium, pp. 819–824, 2018.
25. Tepner S, Ney L, Linse M, Lorenz A, Pospischil M, Clement F. Studying knotless screen patterns for fine-line screen printing of Si-solar cells. *IEEE J. Photovolt*. 2020;10(2):319-325. doi:10.1109/JPHOTOV.2019.2959939
26. Tepner S, Ney L, Linse M, Lorenz A, Pospischil M, Clement F. Knotless screens for ultra-fine line screen printing of Si-solar cells—towards front-side silver electrode widths below 20 μm.

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